

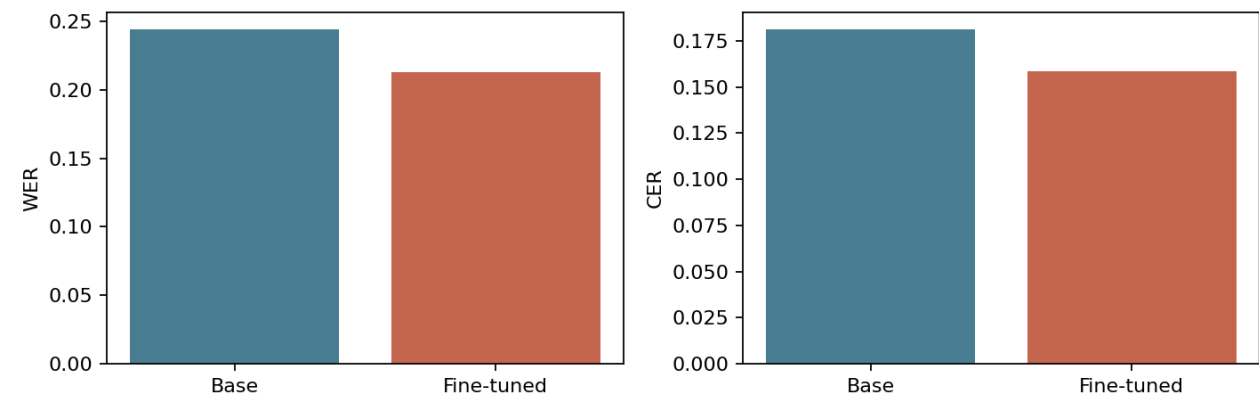
HTX xData | Training Report

Task 4 - Base Parakeet versus selected YCSEP fine-tuned model
Completed E1 held-out evaluation | 12 September 2026

1. Main Result

Fine-tuning improves the full held-out Daily Ketchup (TDK) set: WER falls by **3.1152 percentage points**, a **12.7531% relative reduction**. This is a bounded 10.046-hour adaptation experiment, not training on all available non-TDK audio. All 218,011 metadata rows across 367 videos have paired successful predictions; none were dropped to improve the score.

Metric	Base Parakeet	Fine-tuned E1
Corpus WER	24.4272%	21.3120%
Corpus CER	18.1281%	15.8369%
Word edits / reference words	607,513 / 2,487,035	530,036 / 2,487,035
Character edits / reference characters	1,831,500 / 10,103,081	1,600,016 / 10,103,081
Coverage	218,011 / 218,011	218,011 / 218,011



Uncertainty and Breadth

A source-video cluster bootstrap (2,000 draws; seed 2026) estimates a 95% interval of **[-3.2273, -3.0055] percentage points** for FT-minus-base WER. 355 videos improve and 12 regress. The interval assumes independent videos; repeated hosts and related episodes can violate this assumption. It describes uncertainty within this corpus, not guaranteed performance on other Singapore English recordings.

Evidence: asr-train/results/E1-TDK/comparison.json, base-metrics.json, ft-metrics.json and by-video.csv. The released TDK_subset.csv contains both predictions, explicit error status and per-row WER/CER. Corpus metrics pool edit counts, rather than averaging row or shard WER.

Experimental Design

2. Data, Preprocessing and Selection

The YCSEP static metadata [1] contains 757,072 rows, 1,312 videos and approximately 620.29 annotated segment-hours. All 218,011 TDK rows (172.91 annotated hours) were reserved for final evaluation. Non-TDK source videos were deterministically split using seed 2026. The pilot sampled channels with balanced target hours and spread selection across videos: 10,972 training clips / 10.046 hours from 852 videos; 2,133 validation clips / 2.023 hours from 93 disjoint videos. This is video-disjoint, not proven speaker-disjoint.

Metadata and decoded audio were checked for valid boundaries, usable text, finite nonzero waveforms, duration and sample rate. Audio was converted to mono 16 kHz PCM; transcripts were preserved. **No denoising, VAD trimming, UTMOS/SIGMOS filtering or base-WER-based deletion was used.** Such filters could erase difficult but correct accent/code-switching examples. Valid audio is not proof of correct transcript alignment.

3. Model and Optimization

The specified NVIDIA Parakeet TDT 0.6B v3 model [2] was fine-tuned with PyTorch/NeMo using full-parameter updates, not LoRA or frozen-encoder adaptation. The pretrained 8,192-token SentencePiece tokenizer and acoustic frontend were retained: 128 log-Mel features, 25 ms Hann window, 10 ms hop, 512-point FFT and per-feature normalization. Existing SpecAugment was retained. Replacing the tokenizer would change the output representation without evidence that it is the main bottleneck.

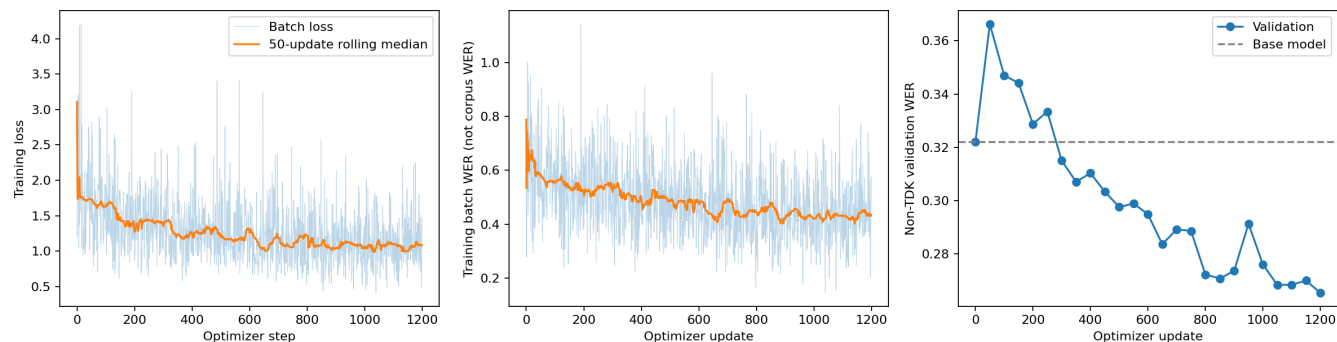
Setting	E1 value / rationale
AdamW	Learning rate 1e-5; betas (0.9, 0.98); weight decay 0.001. Conservative adaptation from pretrained weights.
Memory / stability	BF16; batch 8; accumulate 4 (effective 32); gradient clipping 1.0.
Budget / selection	1,200 updates; 120-minute training cap; validate every 50 updates. Lowest non-TDK validation WER selects checkpoint.
Measured runtime	73.05 minutes on one dedicated A100 80 GB; peak allocated GPU memory 35.60 GB. Runtime excludes audio preparation and evaluation.

Audio Provenance

During the original clip-URL outage, baseline audio was recovered from original YCSEP WAV timestamp ranges and encoded as MP3. Eight Azure shards called the real HTTP /asr endpoint. Fine-tuned evaluation reused the same hash-verified MP3 cache. These inputs are not byte-identical to the original audio-column MP3s; this operational deviation is explicit. The strict merge preserved source order and every original row, verified all eight shards and bound inference to the selected model checksum.

Learning and Error Analysis

4. Training and Validation Curves



Native NeMo validation WER improved from 32.2042% to 26.5402%, with the best checkpoint at update 1,200. An early rise to 36.63% at update 50 was followed by improvement below the base by update 300. Training-loss median declined from about 1.683 in the first 100 updates to 1.058 in the last 100. Late validation fluctuations show that falling batch loss alone is insufficient for selection. The best checkpoint lies at the budget boundary, so convergence or globally optimal hyperparameters are not established. Training batch WER is noisy and is not corpus WER; native validation and normalized TDK WER are not directly interchangeable.

5. Where Errors Remain

Clip duration	Rows	Base WER	FT WER
<2s	117,758	49.81%	43.15%
2-5s	62,338	24.30%	21.48%
5-10s	29,134	16.95%	14.83%
>=10s	8,781	13.66%	11.67%

All duration bands improve, but clips under two seconds remain much harder. At row level, 65,627 improve, 36,169 regress and 116,215 tie in word-edit count. Fragmentation, overlapping speech and annotation alignment are plausible contributors; these mechanisms were not manually verified. Short-reference WER can exceed 100% because insertions are unbounded.

A diagnostic example improves the reference "PIE, ECP, KPE, you know." from separated letters to the correct acronyms. Another reduces repeated filler while still omitting reference words. Conversely, some fine-tuned outputs introduce repeated phrases where the base did not. Thus the aggregate gain is not a guarantee of factual or linguistic correctness for individual clips. The largest-change examples in error-examples.json were selected for diagnosis and are not an unbiased qualitative sample.

Both systems use the same Unicode NFKC, case-folded word normalization with internal apostrophes; CER excludes spaces. Punctuation and capitalization are not evaluated. No significance claim is made for Singlish, names or code-switching subgroups because those categories were not independently labeled.

Next Experiments and Limits

6. Prioritized Improvement Plan

Priority / experiment	Evidence and controlled comparison
1. Alignment audit	Label a stratified non-TDK sample across duration, channels and high/low base-WER cases. Separate bad alignment from difficult but correct speech; correct or exclude only verified defects. Keep the validation set fixed.
2. More representative data	Compare nested 10-hour and 25-hour YCSEP training sets at equal update budgets, then compare at matched data exposure. An E2 25-hour run is separate and not included here. Select on non-TDK validation only.
3. Learning-rate / duration study	Try a bounded 5e-6, 1e-5, 2e-5 grid with the same data and seeds; extend training only if validation supports it. Compare full updates with a frozen-encoder baseline to measure adaptation/memory trade-offs.
4. Context and robustness	On non-TDK data, compare adjacent same-speaker segment concatenation against current short clips, carefully preserving transcript alignment. Ablate denoising or noise augmentation one at a time; retain an unprocessed control.
5. External Singapore English	After access/license review, sample IMDA National Speech Corpus [3]. Compare YCSEP-only with a bounded NSC mixture, controlling total hours and updates. Hold out distinct source recordings and audit cross-dataset duplication.
6. Generalization check	Use multiple seeds and a new untouched channel/speaker-disjoint evaluation set. This TDK test has now been examined; do not repeatedly select improvements against it. Report WER/CER plus coverage and operational throughput.

Interpretation Limits

The experiment establishes a useful corpus-level gain with modest adaptation data, not optimal preprocessing or the best possible model. References may contain annotation errors; related videos and shared speakers weaken independence. Channel-balanced training is not identical to the natural frequency distribution. The recovered-audio path, one training seed and one selected configuration limit generalization claims. No broader robustness benefit from denoising or quality filtering can be claimed because those ablations were not run.

References and Evidence

[1] YCSEP, Harvard Dataverse, DOI 10.7910/DVN/B7JRID. Source: YCSEP_static.csv; source SHA-256 is recorded in merge provenance.
[2] NVIDIA, [Parakeet TDT 0.6B v3 model card](#). Model architecture and intended audio inputs; experiment-specific configuration is in results/E1/model-config.yaml.
[3] IMDA, [National Speech Corpus](#). Proposed external data only; not used in E1.
Recorded experiment evidence: results/E1/training-result.json and metrics.csv; paired evaluation and reproducible analysis helpers under asr-train/. External pages checked 12 September 2026.